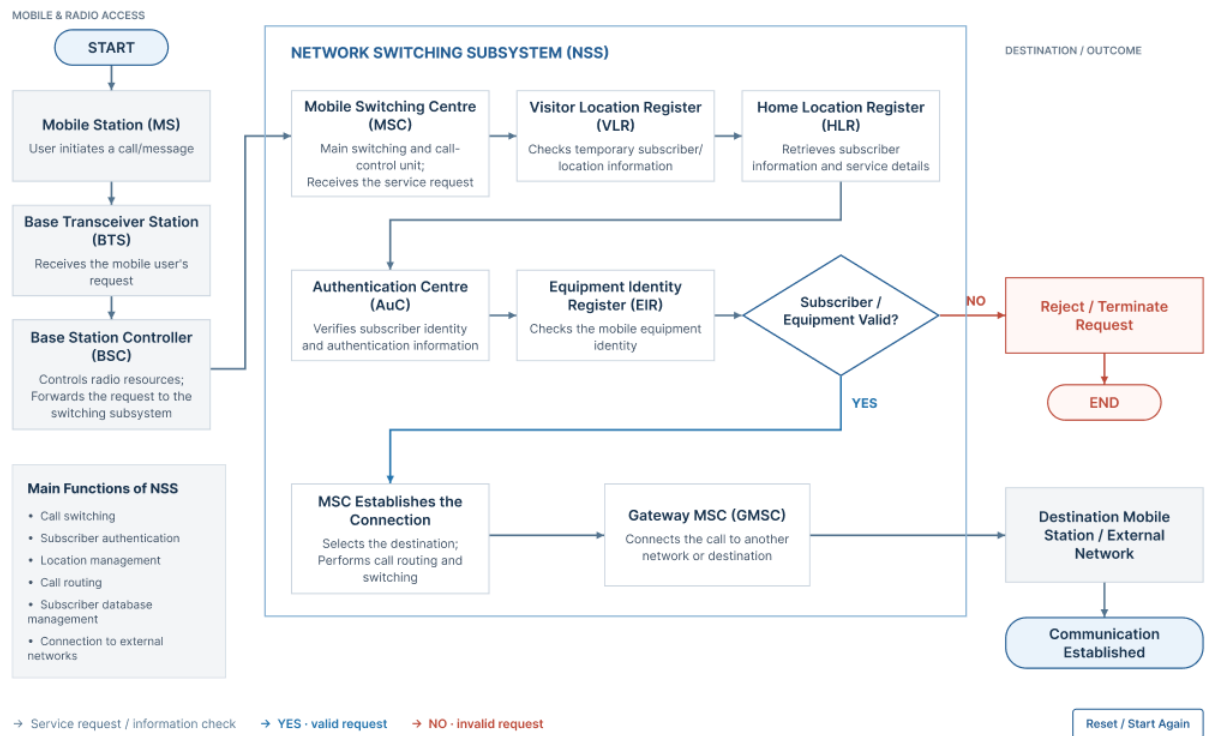


Operation of Network Switching Subsystem (NSS)

Call and Communication Flow in a Mobile Cellular System



MSC — component information

NETWORK SWITCHING SUBSYSTEM (NSS)

MSC

Mobile Switching Centre

Role in the network

The main switching and call-control unit. It receives service requests from the BSC and coordinates the checks required to serve the subscriber.

Place in the illustrated flow

First NSS component after the BSC. After the validity decision is YES, the MSC selects the destination and performs call routing and switching.

BSC → MSC → VLR

The "MSC Establishes the Connection" process is a function of the MSC, not a separate network component.

[Back to Flowchart](#)Component reference

HLR — component information

NETWORK SWITCHING SUBSYSTEM (NSS)

HLR

Home Location Register

Role in the network

The permanent subscriber database. It stores subscription details, permitted services and information used to locate the subscriber's serving network.

Place in the illustrated flow

Shown after the VLR. Subscriber information and service details are retrieved before authentication proceeds at the AuC.

VLR → HLR → AuC

The HLR holds the home subscriber record; the VLR holds temporary information for the serving area.

[Back to Flowchart](#)Component reference

VLR — component information

NETWORK SWITCHING SUBSYSTEM (NSS)

VLR

Visitor Location Register

Role in the network

A temporary subscriber database associated with the serving MSC. It stores information about subscribers currently registered in its service area.

Place in the illustrated flow

Shown after the MSC. Temporary subscriber and location information is checked before the illustrated flow retrieves the HLR record.

MSC → VLR → HLR

Locally available subscriber data supports call handling and reduces repeated requests to the home database.

[Back to Flowchart](#)Component reference

AuC — component information

NETWORK SWITCHING SUBSYSTEM (NSS)

AuC

Authentication Centre

Role in the network

Provides authentication information derived from subscriber security credentials. It supports the network in verifying the subscriber's identity.

Place in the illustrated flow

Shown after the HLR. Subscriber authentication is verified before the equipment identity check at the EIR.

HLR → AuC → EIR

Authentication concerns the subscriber identity. Equipment validation is handled separately by the EIR.

[Back to Flowchart](#)Component reference

EIR — component information

NETWORK SWITCHING SUBSYSTEM (NSS)

EIR

Equipment Identity Register

Role in the network

Maintains mobile equipment identity records, typically using the IMEI. It allows the network to check whether a device is permitted, flagged or barred.

Place in the illustrated flow

Shown after the AuC and immediately before "Subscriber / Equipment Valid?". An invalid request follows the NO path to rejection and END.

AuC → EIR → Validity decision

Equipment identity checking is separate from subscriber authentication; a valid subscriber can use barred equipment.

[Back to Flowchart](#)Component reference

GMSC — component information

NETWORK SWITCHING SUBSYSTEM (NSS)

GMSC

Gateway MSC

Role in the network

Provides the gateway between the mobile network and other networks. It supports routing the call to the appropriate destination network.

Place in the illustrated flow

Shown after "MSC Establishes the Connection" on the YES path. The call is connected onward to the destination mobile station or external network.

MSC connection → GMSC → Destination

The GMSC belongs inside the NSS boundary. The destination mobile station or external network remains outside it.

[Back to Flowchart](#)Component reference